UNIVERSITY OF BIRMINGHAM

School of Mathematics

Programmes in the School of Mathematics Second Examination
Programmes involving Mathematics Second Examination

2LA 06 15552 Level I

Linear Algebra

January Examinations 2021-22
One Hour and Thirty Minutes

Full marks will be obtained with complete answers to BOTH questions. Each question carries equal weight. You are advised to initially spend no more than 45 minutes on each question and then to return to any incomplete questions if you have time at the end. An indication of the number of marks allocated to parts of questions is shown in square brackets.

No calculator is permitted in this examination.

Section A

1. (a) Explaining your working and quoting standard results where required, find the rank and nullity of the following matrices.

$$X = \begin{pmatrix} 1 & 2 & -3 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{pmatrix} \qquad Y = \begin{pmatrix} 1 & 1 & -1 & 1 \\ 1 & 2 & 0 & 3 \\ 2 & 3 & -1 & 4 \\ 3 & 4 & -2 & 5 \end{pmatrix}.$$

[5]

(b) For each of the following matrices, say whether it is similar to a diagonal matrix over the reals. Justify your answers.

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix} \qquad B = \begin{pmatrix} -13 & -28 \\ 6 & 13 \end{pmatrix} \qquad C = \begin{pmatrix} -7 & -5 \\ 10 & 7 \end{pmatrix}.$$

[12]

(c) Write the symmetric bilinear form

$$F\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) = x_1y_1 + 2x_1y_2 + 2x_2y_1 - x_2y_2$$

on the space V of 2×1 column vectors over $\mathbb{R}$ in matrix form with respect to the usual basis e_1, e_2. Find a vector e_2' such that $\{e_1, e_2'\}$ is linearly independent and $F(e_1, e_2') = 0$. For your e_2' find $F(e_2', e_2')$ and hence write down the matrix representing F with respect to e_1, e_2'. Is F an inner product? [8]

Section B

2. In the following, $V = \mathbb{R}^5_{\text{col}}$ is the set of 5×1 (column) matrices over $\mathbb{R}$, and the matrix A is a 5×5 matrix over $\mathbb{R}$ considered as a map $V \to V$ defined by $v \mapsto Av$.

(a) Say what it means for $v \in V$ to be an eigenvector of A with corresponding eigenvalue λ. [2]

(b) If v_1, v_2, v_3, v_4, v_5 is a basis of V where each v_i is an eigenvector with eigenvalue λ_i for $i = 1 \ldots 5$, write down the matrix representation of A with respect to this basis. [2]

(c) Show that the minimum polynomial $m(x)$ of the matrix

$$A = \begin{pmatrix} 2 & 0 & 0 & 3 & 0 \\ 3 & 2 & 0 & 3 & 3 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ -3 & 0 & 0 & -3 & -1 \end{pmatrix}$$

has degree 2 and say what $m(x)$ is. [8]

(d) Find a basis of eigenvectors for the matrix A above. Say what the eigenvalues corresponding to each of your eigenvectors are. [10]

(e) Hence write down a matrix P such that $P^{-1}AP = D$ where D is diagonal, and say what your D is. [3]

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats and outer-wear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) **MUST** be placed in the designated area.
- Check that you **DO NOT** have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **MUST** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are **NOT** permitted to use a mobile phone as a clock. If you have difficulty in seeing a clock, please alert an Invigilator.
- You are **NOT** permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately.
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to the Student Conduct procedures.